PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

School of Information Science

Department of Master of Computer Applications

One - Stop Personalized Career & Education Advisor Software

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BONAFIDE CERTIFICATE

Certified that this project titled "**One - Stop Personalized Career & Education Advisor Software** " is a bonafide work of " **SHARANBIR (20242MCA0118)**", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of **Master of Computer Applications** during 2025-26.

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DECLARATION

We the students of first year Master of Computer Applications at Presidency University, Bengaluru, named **SHARANBIR**, hereby declare that the major project titled **"One-Stop Personalized Career & Education Advisor Software"** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of Master of Computer Applications during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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SHARANBIR

Abstract

Choosing the right career path has become increasingly challenging in today's rapidly evolving job market, where new technologies and roles emerge continuously. Many students and professionals struggle to identify suitable career options due to a lack of proper guidance, limited awareness of industry requirements, and mismatch between their skills and job expectations. Traditional career counseling methods are often time-consuming, less personalized, and not easily accessible to everyone. This creates a need for an intelligent system that can provide personalized, data-driven career guidance to users.

The CareerShala system, titled *"**One-Stop Personalized Career & Education Advisor Software**,"* is designed to address these challenges by leveraging Artificial Intelligence (AI) and Machine Learning (ML) techniques. The system collects user data such as educational background, skills, resume content, and career preferences, and processes it using advanced analytical methods. Natural Language Processing (NLP) is used to extract meaningful information from resumes, while machine learning models are applied to generate career recommendations and salary predictions. The platform integrates multiple modules, including resume analysis, skill gap detection, career assessment, and an AI-based chatbot advisor, all within a unified interface.

The system further enhances decision-making by identifying skill gaps and suggesting appropriate learning paths to improve employability. It also provides insights into salary trends and career growth opportunities, helping users understand their potential in different domains. The interactive user interface ensures that complex data is presented in a simple and intuitive manner, making it accessible even to non-technical users. By combining multiple features into a single platform, the system eliminates the need for users to rely on multiple tools for career planning.

The results of the CareerShala system demonstrate its effectiveness in providing accurate and personalized career guidance. The implemented modules successfully analyze user inputs and generate meaningful insights, enabling users to make informed career decisions. Overall, the platform serves as a scalable and efficient solution for modern career guidance, with the potential for further enhancements through integration of real-time data and advanced AI models.

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List of Abbreviations

Abbreviation	**Full Form**
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
ATS	Applicant Tracking System
UI	User Interface
UX	User Experience
API	Application Programming Interface
DB	Database
ER	Entity Relationship
DFD	Data Flow Diagram
SDLC	Software Development Life Cycle
IoTWF	Internet of Things World Forum
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
CSV	Comma Separated Values
IDE	Integrated Development Environment
KPI	Key Performance Indicator
GUI	Graphical User Interface
CPU	Central Processing Unit
RAM	Random Access Memory
PDF	Portable Document Format
DOCX	Microsoft Word Document Format

Chapter 1

Introduction

1.1 Background

In recent years, the rapid growth of technology and digital transformation has significantly changed the nature of career opportunities across industries. Students and job seekers are now exposed to a wide variety of career paths, making decision-making more complex than ever before. Traditional career guidance methods rely on manual counseling, which is often limited, time-consuming, and lacks personalization.With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), intelligent systems can now analyze large amounts of data and provide personalized recommendations. These systems can evaluate user profiles, skills, educational background, and industry trends to suggest suitable career paths.

Several studies have shown that AI-based career guidance systems improve decision-making accuracy and provide better career alignment compared to traditional methods [1], [2]. However, most existing solutions lack integration and fail to provide a complete end-to-end career advisory system.

The proposed **One-Stop Personalized Career & Education Advisor Software (CareerShala)** aims to address these challenges by integrating multiple intelligent modules such as resume analysis, skill gap detection, career recommendation, and salary prediction into a single unified platform.

1.2 Statistics

The need for an intelligent career guidance system is supported by several global and regional statistics. According to industry reports, a significant percentage of students are uncertain about their career choices due to lack of proper guidance. Studies indicate that more than 60% of graduates feel their education does not align with job requirements, highlighting a major skill gap in the workforce [3].

In India, employability reports suggest that only around 45% of graduates are considered job-ready, mainly due to lack of practical skills and awareness of industry demands [4].

Additionally, surveys show that many students choose career paths based on peer influence rather than informed decision-making.

The increasing demand for skilled professionals in fields such as Artificial Intelligence, Data Science, and Software Development further emphasizes the need for systems that can guide users effectively. These statistics clearly indicate the importance of developing a personalized and intelligent career advisory platform.

1.3 Prior existing technologies

Several technologies and platforms currently exist to assist users in career planning and skill development. Popular platforms such as LinkedIn Learning, Coursera, and Glassdoor provide course recommendations, job listings, and salary insights. However, these systems operate independently and lack integration.

AI-based career guidance systems have been proposed in recent research. For example, AI-powered chatbots use Natural Language Processing (NLP) to interact with users and provide basic career suggestions [5]. Machine learning-based recommendation systems analyze user data to suggest career paths based on similarity and historical patterns [2].

Resume analysis tools use NLP techniques to extract skills and match them with job descriptions. However, most of these tools focus only on specific functionalities and do not provide a comprehensive solution that includes skill gap analysis, career simulation, and predictive insights.

Therefore, there is a need for an integrated system that combines multiple technologies into a single platform, providing a complete career advisory solution.

1.4 Proposed approach

Aim of the Project

The main aim of this project is to develop an AI-based system that provides personalized career and education guidance by analyzing user data and industry trends.

Motivation

The motivation behind this project is to address the challenges faced by students in choosing the right career path. The lack of proper guidance, awareness of required skills, and real-time industry insights often leads to poor decision-making. This project aims to bridge the gap between education and employment by providing intelligent recommendations.

Proposed Approach

The CareerShala system adopts a multi-module architecture that integrates various AI-driven components. The system collects user data such as profile information, skills, and resumes. Using NLP techniques, the system extracts relevant information and processes it for analysis.

The Skill Gap Analysis module compares user skills with industry requirements to identify missing competencies. The Career Recommendation Engine suggests suitable career paths based on user data and market trends. The Salary Prediction module estimates expected salary based on skills and experience.

Additionally, an AI Career Advisor chatbot provides real-time interaction and guidance. The system also includes a Career Sandbox module that allows users to simulate different career scenarios.

Applications of the Project

- Career guidance for students and job seekers
- Resume evaluation and skill improvement
- Interview preparation and assessment
- Career planning and decision-making

Limitations of the Proposed Approach

- Depends on the accuracy of user-provided data
- Predictions may vary based on dataset quality
- Limited real-time job integration in current version
- AI models require continuous updates for accuracy

1.5 Objectives

The objectives of the CareerShala system are designed to cover multiple aspects of system development:

i. Behaviour:

To design an interactive system that provides real-time career guidance through an AI chatbot and user-friendly interface.

ii. Analysis:

To analyze user profiles and resumes using NLP techniques for extracting skills and identifying career opportunities.

iii. System Management:

To develop a scalable system architecture that efficiently integrates multiple modules such as recommendation, prediction, and analysis.

iv. Security:

To ensure secure handling of user data through authentication and data protection mechanisms.

v. Deployment:

To deploy the system as a web-based platform accessible to users across different devices and environments

1.6 SDGs

The CareerShala project aligns with several United Nations Sustainable Development Goals (SDGs).

SDG 4 – Quality Education: The system promotes personalized learning by recommending courses and skill development paths.

SDG 8 – Decent Work and Economic Growth: It helps users identify suitable career opportunities and improve employability.

SDG 9 – Industry, Innovation, and Infrastructure: The use of AI technologies supports innovation in career guidance systems.

These alignments demonstrate how the project contributes to global development goals by improving education and employment opportunities [6].

Fig 1.6 Sustainable development goals

1.7 Overview of project report

This project report is systematically organized into multiple chapters, each focusing on a specific aspect of the development and implementation of the **CareerShala system**. The structure of the report ensures a logical flow from the introduction of the problem to the final conclusions and future enhancements.

Chapter 1 provides an introduction to the project, presenting the background, problem statement, motivation, objectives, and proposed approach. It highlights the need for an intelligent career guidance system and explains how the CareerShala platform addresses existing challenges. Additionally, it outlines the alignment of the project with Sustainable Development Goals (SDGs) and provides an overview of the entire report.

Chapter 2 focuses on the literature review, analyzing existing research and technologies related to career recommendation systems, AI-based advisory tools, and resume analysis systems. This chapter compares different approaches, identifies their strengths and limitations, and establishes the research gap that the proposed system aims to address.

Chapter 3 describes the methodology adopted for the development of the system. It explains the development model, system workflow, and the techniques used for data processing, analysis, and implementation. This chapter also discusses how Agile methodology is applied to ensure iterative development and continuous improvement.

Chapter 4 presents project management aspects, including planning, scheduling, and resource allocation. It covers tools such as Gantt charts and risk management strategies, ensuring that the project is executed efficiently within the given timeline.

Chapter 5 discusses system analysis and design in detail. It includes system architecture, data flow diagrams, use case diagrams, and flowcharts. This chapter explains how different modules interact with each other and how data flows through the system to generate outputs.

Chapter 6 provides implementation details, including the technologies used for frontend, backend, and database development. It explains how various modules such as resume analysis, skill gap detection, and career recommendation are implemented and integrated into the system.

Chapter 7 presents the evaluation and results of the system. It includes performance analysis, system outputs, and comparison with existing approaches. This chapter demonstrates the effectiveness of the proposed system in providing accurate and personalized career guidance.

Chapter 8 discusses social, ethical, and sustainability aspects related to the system. It highlights issues such as data privacy, security, and responsible use of AI technologies, along with their impact on society.

Finally, **Chapter 9** concludes the report by summarizing the key contributions of the project and discussing future enhancements. It outlines potential improvements such as real-time job integration, advanced AI models, and expansion of system capabilities.
Overall, this report provides a comprehensive understanding of the design, development, and evaluation of the CareerShala system, demonstrating its potential as an effective solution for personalized career and education guidanc

Chapter 2
Literature review

2.1 Literature Review

Bhaskarwar et al. [1] proposed an AI-based career counseling system designed to assist students in selecting appropriate career paths based on their interests, academic performance, and skill sets. The system utilizes machine learning algorithms to analyze user inputs and generate personalized career recommendations. The study demonstrated that AI-driven systems can significantly improve decision-making accuracy compared to traditional counseling approaches. However, the system primarily focuses on static user data and lacks integration with real-time job market trends. Additionally, it does not incorporate modules such as resume analysis or predictive modeling, limiting its overall effectiveness. The authors suggested that future improvements could include dynamic data integration and enhancement of recommendation accuracy using advanced AI techniques.

Siswipraptini et al. [2] developed a personalized career-path recommendation model using data mining techniques for Information Technology students. The approach involves analyzing academic records, skill sets, and preferences to predict suitable career options. The system achieved improved recommendation accuracy by utilizing classification algorithms and pattern recognition methods. Despite its effectiveness, the model does not consider external factors such as industry demand and job trends, which are crucial for real-world applications. Furthermore, the system lacks adaptability to changing user profiles. Future work suggested includes incorporating real-time data sources and enhancing the model with adaptive learning techniques.

Supriyanto et al. [3] introduced an expert system for career guidance based on rule-based decision-making. The system simulates human expertise by using predefined rules to recommend career options to students. While the system provides consistent outputs, it is limited by its inability to learn from new data or adapt to dynamic changes in user preferences and industry requirements. The lack of machine learning capabilities restricts its scalability and flexibility. The authors recommended integrating AI-based techniques to improve system adaptability and performance.

Roy and Sharma [4] proposed an AI-powered career guidance framework that uses machine learning models to provide scalable and personalized recommendations. The system analyzes multiple user attributes, including skills, interests, and educational background, to generate career suggestions. The results indicated improved performance compared to traditional systems. However, the framework does not include resume-based analysis or skill gap identification, which are essential for comprehensive career planning. The study suggests enhancing the system by integrating additional modules such as predictive analytics and real-time job data.

Uthayam et al. [5] developed an AI-based career advisor system that incorporates Natural Language Processing (NLP) techniques for analyzing user input and resumes. The system extracts key information from resumes and provides suggestions for improving employability. The approach demonstrated the effectiveness of NLP in handling unstructured data. However, the system is limited in scope as it focuses mainly on resume analysis and does not provide complete career recommendations or learning pathways. Future work includes expanding the system to include skill gap analysis and recommendation engines.

Thakare et al. [6] presented an AI-driven career advisor system that utilizes machine learning algorithms to match user profiles with suitable career options. The system provides personalized recommendations based on user inputs such as skills and interests. While the system improves decision-making, it lacks integration with external datasets such as job market trends and salary information. Additionally, the absence of interactive features such as chatbots limits user engagement. The authors suggested incorporating real-time data and interactive modules to enhance system usability.

Nithin et al. [7] proposed a one-stop personalized career and educational advisor system that integrates multiple features into a single platform. The system combines career recommendations and educational guidance, providing a more comprehensive solution compared to earlier approaches. However, the system does not utilize advanced AI techniques such as predictive modeling or NLP-based analysis. This limits its ability to provide highly accurate and adaptive recommendations. Future improvements suggested include incorporating AI-driven modules and enhancing system scalability.
Ahmed et al. [8] explored the use of metaheuristic optimization techniques for improving machine learning model performance. The study demonstrated that optimization algorithms can enhance prediction accuracy by fine-tuning model parameters. Although the research is focused on medical image analysis, the concept of optimization can be applied to career recommendation systems to improve performance. The limitation of the study is its domain-specific application, which requires adaptation for use in career guidance systems.

Liu et al. [9] introduced transformer-based architectures for improved data representation and analysis.

The study highlighted the advantages of using advanced deep learning models for capturing both local and global features. While the research is primarily focused on image processing, the methodology can be extended to NLP-based systems for resume analysis and chatbot interaction. However, the complexity and computational requirements of such models may limit their practical implementation in large-scale systems.

Zaidi et al. [10] proposed a two-stage framework for classification and prediction tasks, combining feature extraction and decision-making processes. The study demonstrated improved accuracy by separating detection and classification stages. This approach can be applied to career guidance systems by separating skill extraction and recommendation modules. However, the framework increases system complexity and requires efficient integration techniques.

Table 2.1 Summary of Literature Reviews

Ref	Approach	Method Used	Advantages	Limitations
[1]	AI Career System	ML	Personalized	No real-time data
[2]	Recommendation Model	Data Mining	Accurate	No industry trends
[3]	Expert System	Rule-based	Consistent	Not adaptive
[4]	AI Framework	ML Models	Scalable	No resume analysis
[5]	Resume Analyzer	NLP	Skill extraction	Limited scope
[6]	AI Advisor	ML	Personalized	No chatbot
[7]	Integrated System	Basic AI	Multi-feature	No advanced AI
[8]	Optimization Model	Metaheuristics	High accuracy	Domain-specific
[9]	Transformer Model	Deep Learning	Powerful analysis	High complexity
[10]	Hybrid Framework	Multi-stage	Better accuracy	Complex system

2.2 Summary of Literatures Reviewed

The literature review highlights the growing importance of AI-based systems in career guidance. Most studies focus on improving recommendation accuracy using machine learning and data mining techniques. Some research emphasizes the use of NLP for resume analysis, while others explore expert systems and optimization techniques.

However, several limitations are identified across existing systems:

- Lack of integration between different modules
- Limited use of real-time data

- Absence of predictive analytics
- Lack of personalization in some systems
- High system complexity in advanced models

The proposed CareerShala system addresses these gaps by integrating multiple modules into a single platform and using AI techniques for improved accuracy and adaptability.

Chapter 3
Methodology

3.1 Introduction

The development of the **CareerShala system** requires a structured and flexible methodology to ensure efficient design, implementation, and testing. Considering the dynamic nature of the project and the need for continuous improvement, an **Agile-based development approach combined with the V-Model for testing** is adopted.

The Agile model supports iterative development and allows continuous feedback, while the V-Model ensures proper verification and validation at each stage of development. This hybrid approach ensures both flexibility and quality assurance in the system.

3.2 Selected Methodology: Agile

The Agile methodology focuses on iterative development, where the system is built in small modules and continuously improved. Each module of the CareerShala system, such as resume analysis, skill gap detection, and career recommendation, is developed and tested independently.

The V-Model complements Agile by providing a structured testing framework. Each development phase has a corresponding testing phase, ensuring early detection of errors and improved system reliability.

Fig 3.2 Agile Model

3.3 V-Model Methodology

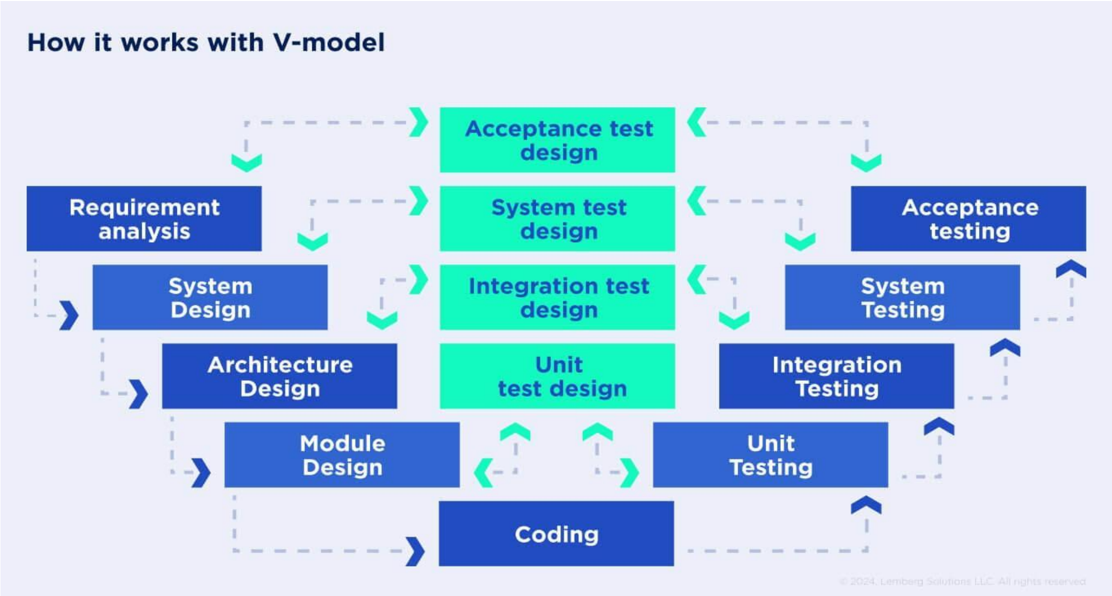

Fig 3.3 V-Model Methodology

Explanation:
The V-Model represents the software development lifecycle in a "V" shape, where the left side represents development phases and the right side represents testing phases.

Verification Phase (Left Side):

- Requirement Analysis
- System Design
- Architecture Design
- Module Design

Validation Phase (Right Side):

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing

This model ensures that each stage is verified before moving to the next stage, reducing errors and improving system quality.

3.5 Agile Development Process

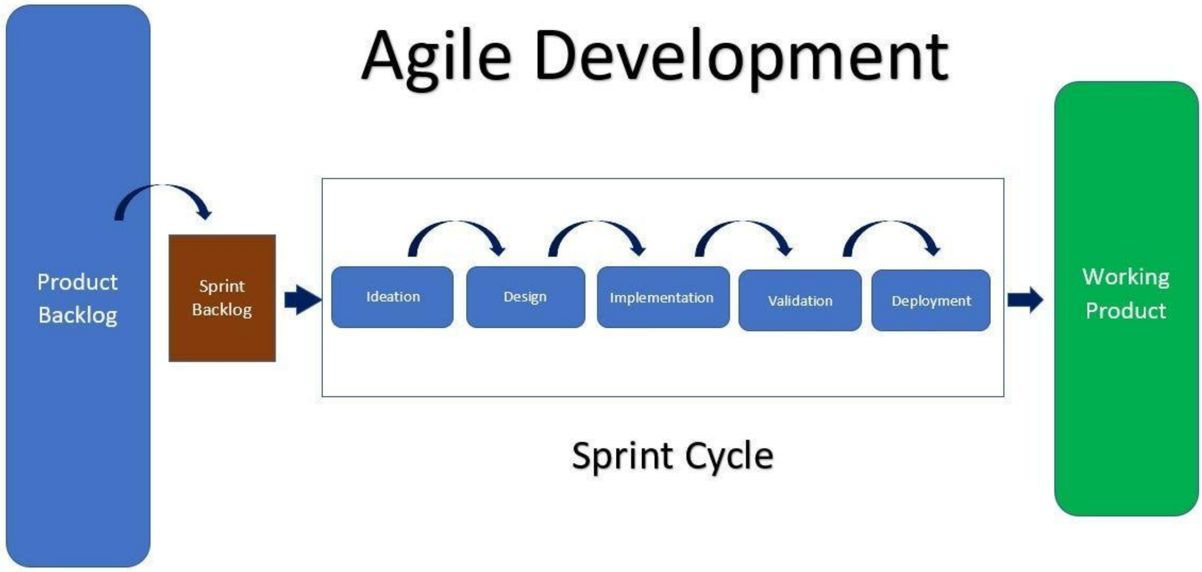

Fig 3.5 Agile Methodology Workflow

Explanation:

The Agile model divides the project into multiple iterations (sprints):

- Requirement gathering
- Development
- Testing
- Feedback
- Improvement

Each module of CareerShala is developed in iterations, ensuring continuous enhancement.

3.6 Mapping Project Phases to Methodology

The CareerShala system development is mapped to the methodology stages as follows:

1. Requirement Phase

- Collect user requirements
- Study literature (Chapter 2)
- Identify system features

2. System Design Phase

- Design system architecture
- Define modules (AI chatbot, resume analyzer)
- Create diagrams (DFD, flowchart)

3. Functional Design Phase

- Define functionality of each module
- Design user interface and workflows

4. Unit Design (Software + AI)

- Develop frontend (React)
- Develop backend (Python)
- Implement AI modules (NLP, ML)

5. Unit Testing

- Test each module individually
- Debug errors

6. Integration Testing

- Combine modules
- Ensure proper data flow

7. System Testing

- Test complete system functionality
- Validate outputs

8. User Acceptance Testing

- Verify system with user requirements
- Collect feedback

9. Deployment (Go Live)

- Deploy system
- Make it accessible to users

3.7 Algorithm for CareerShala System

Algorithm: Career Recommendation Process

Step 1: Start

Step 2: Collect user profile data

Step 3: Upload and process resume using NLP

Step 4: Extract skills from resume

Step 5: Fetch job market and course data

Step 6: Perform skill gap analysis

Step 7: Integrate features into decision engine

Step 8: Generate career recommendations

Step 9: Predict salary and job trends

Step 10: Display results to user

Step 11: End

3.8 System Workflow

Fig 3.8 Smart System Architecture

The CareerShala System Architecture integrates AI Career Advisor, Skill Gap Analysis, and Salary Prediction modules with profile, resume, and job market data to generate intelligent career recommendations.

It uses feature integration, NLP-based resume analysis, and simulation modules to provide personalized career paths, learning plans, salary insights, and mock interview support.

Explanation:

The workflow begins with user input and resume upload. The system extracts skills using NLP techniques and classifies them. It then compares user skills with job requirements to identify gaps. The system generates recommendations, predicts salary, and provides career insights. This ensures a complete end-to-end guidance system.

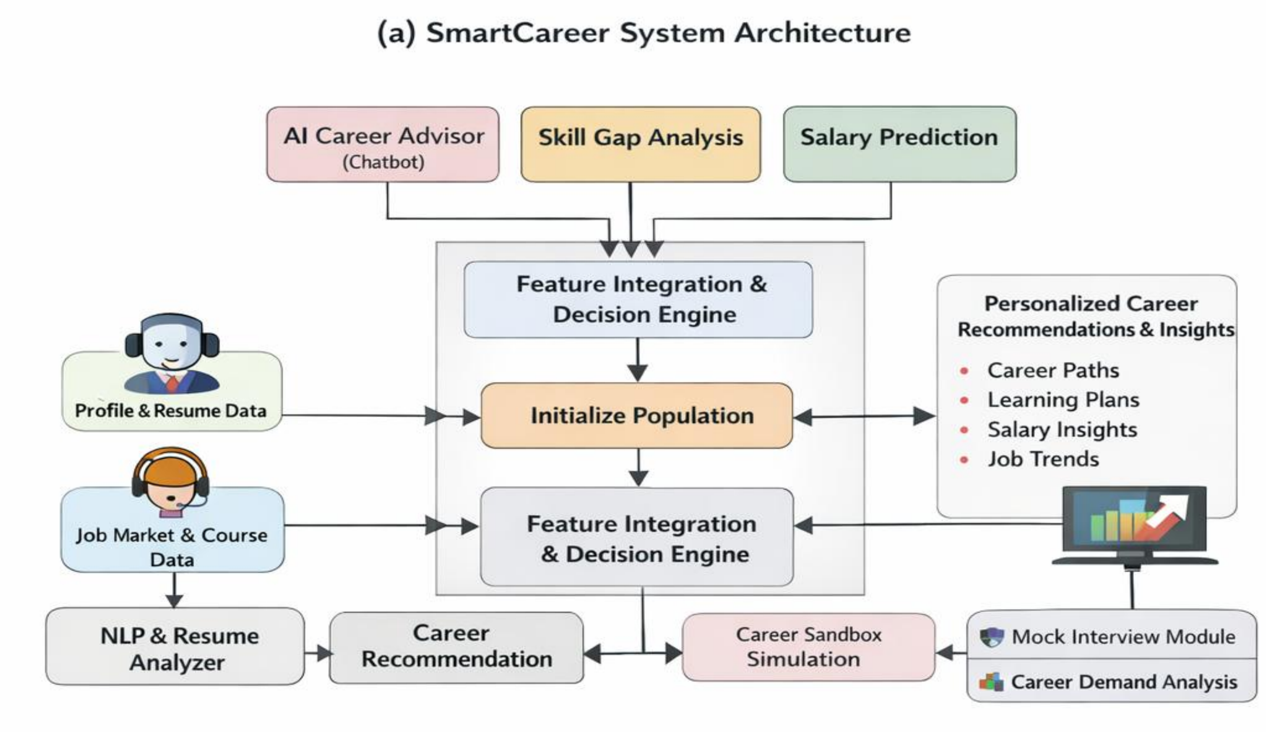

Fig 3.8 Smart System Architecture

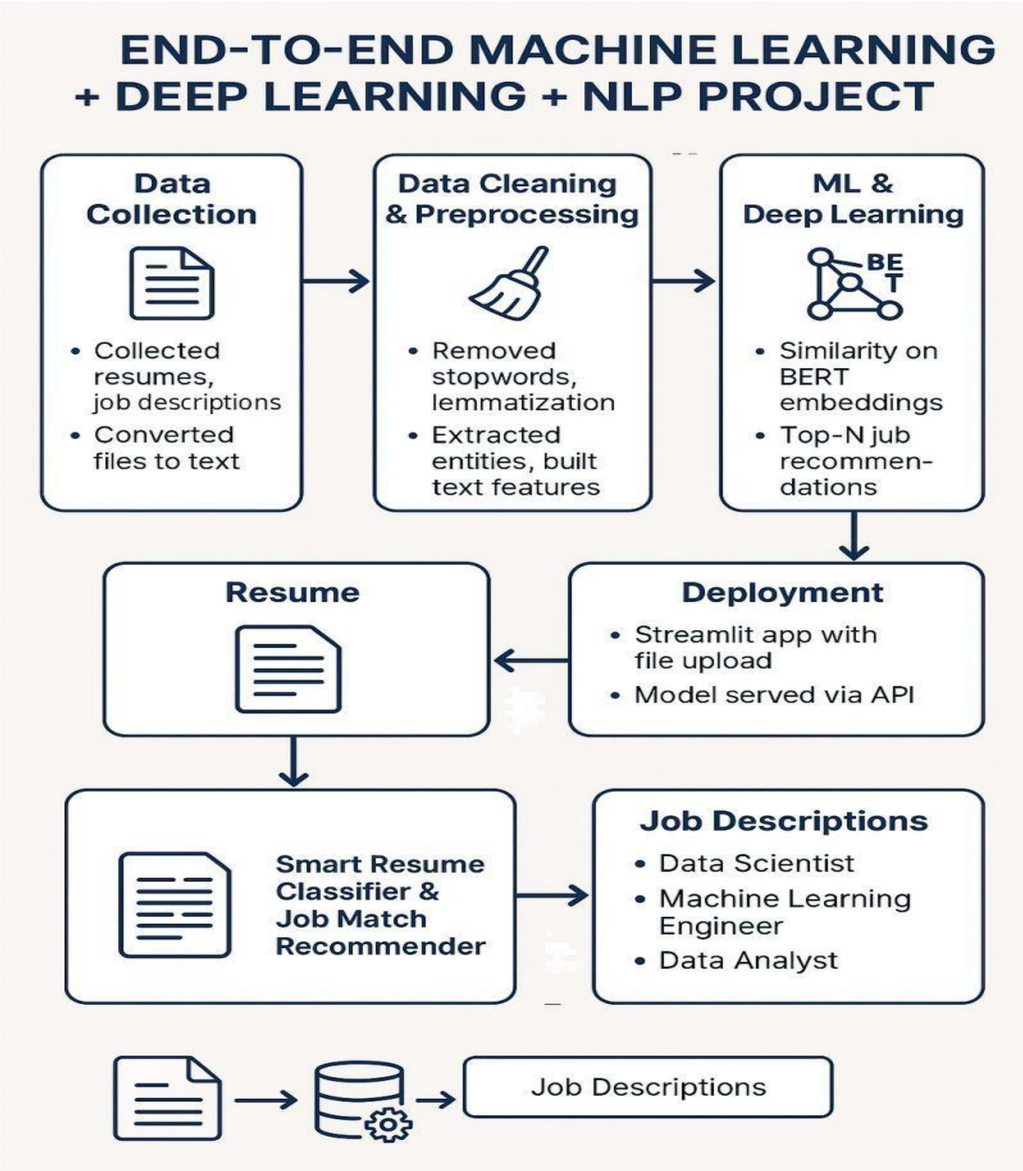

Fig 3. 8.1 CareerShala System Workflow

3.9 Tools Used for Design

- **Draw.io / Lucidchart** → Diagram creation
- **Figma** → UI design
- **GitHub** → Version control
- **VS Code** → Development

3.10 Summary

This chapter explained the methodology used for developing the CareerShala system. The combination of Agile and V-Model ensures both flexibility and reliability. The structured approach helps in efficient development, testing, and deployment of the system.

CHAPTER 4
PROJECT MANAGEMENT

4.1 Project Timeline

Project management is an important phase in the successful development of the CareerShala system. A structured timeline helps in organizing project activities, tracking progress, identifying milestones, and ensuring that the project is completed within the planned schedule.

To manage the CareerShala project effectively, a **Gantt Chart** was used as a project management tool. A Gantt Chart is a visual representation of project activities displayed over a timeline. It illustrates tasks, start dates, end dates, duration, dependencies, milestones, and progress. This helps in monitoring the overall workflow and maintaining coordination between different development stages.

The CareerShala project was divided into two major phases:

- Project Planning Phase
- Project Implementation Phase

The planning phase focused on requirement analysis, literature review, and system design, while the implementation phase included development, testing, deployment, and documentation.

Table 4.1 Project Planning Timeline

Sl. No	Activity	Duration	Start Date	End Date
1	Problem Identification	2 Days	Jan 07	Jan 09
2	Requirement Gathering	2 Days	Jan 09	Jan 11
3	Literature Review	9 Days	Jan 11	Jan 19
4	Feasibility Analysis	5 Days	Jan 20	Jan 24
5	System Architecture Design	7 Days	Jan 25	Jan 31
6	Module Planning	5 Days	Feb 01	Feb 05
7	Proposal Preparation	5 Days	Feb 06	Feb 10

Description

Table 4.1 summarizes the timeline during the project planning phase. The planning activities included identifying project objectives, studying existing literature, analyzing system feasibility, and preparing the project proposal. These activities ensured that the project requirements and implementation strategy were

clearly defined before development began. The timeline was suitable for the CareerShala project as it provided sufficient time for requirement analysis and system planning.

Description

Table 4.2 summarizes the timeline during the project implementation phase. The implementation activities were divided into multiple stages, including frontend and backend development, AI module integration, testing, deployment, and documentation. The final deployment of the CareerShala system was completed on **30th April**, while the documentation phase continued until **7th May**. This timeline was suitable because it allowed adequate time for iterative development, testing, debugging, and final report preparation.

Table 4.2 Gantt Chart Table

Sl. No	Activity	Duration	Start Date	End Date
1	Frontend UI Development	10 Days	Feb 11	Feb 20
2	Backend API Development	10 Days	Feb 21	Mar 02
3	Database Integration	5 Days	Mar 03	Mar 07
4	Resume Analyzer Module	8 Days	Mar 08	Mar 15
5	Skill Gap Analysis Module	7 Days	Mar 16	Mar 22
6	Career Recommendation Engine	7 Days	Mar 23	Mar 29
7	Salary Prediction Module	5 Days	Mar 30	Apr 03
8	AI Chatbot Integration	7 Days	Apr 04	Apr 10
9	Testing and Debugging	8 Days	Apr 11	Apr 18

10	System Integration	5 Days	Apr 19	Apr 23
11	Deployment	7 Days	Apr 24	Apr 30
12	Documentation Preparation	7 Days	May 01	May 07

Description

Figure 4.1 illustrates the complete project schedule of the CareerShala system using a Gantt Chart. The chart visually represents task durations, milestones, dependencies, and deadlines across both planning and implementation phases. It helps in tracking progress, managing resources, and ensuring timely completion of project activities.

Gantt Chart

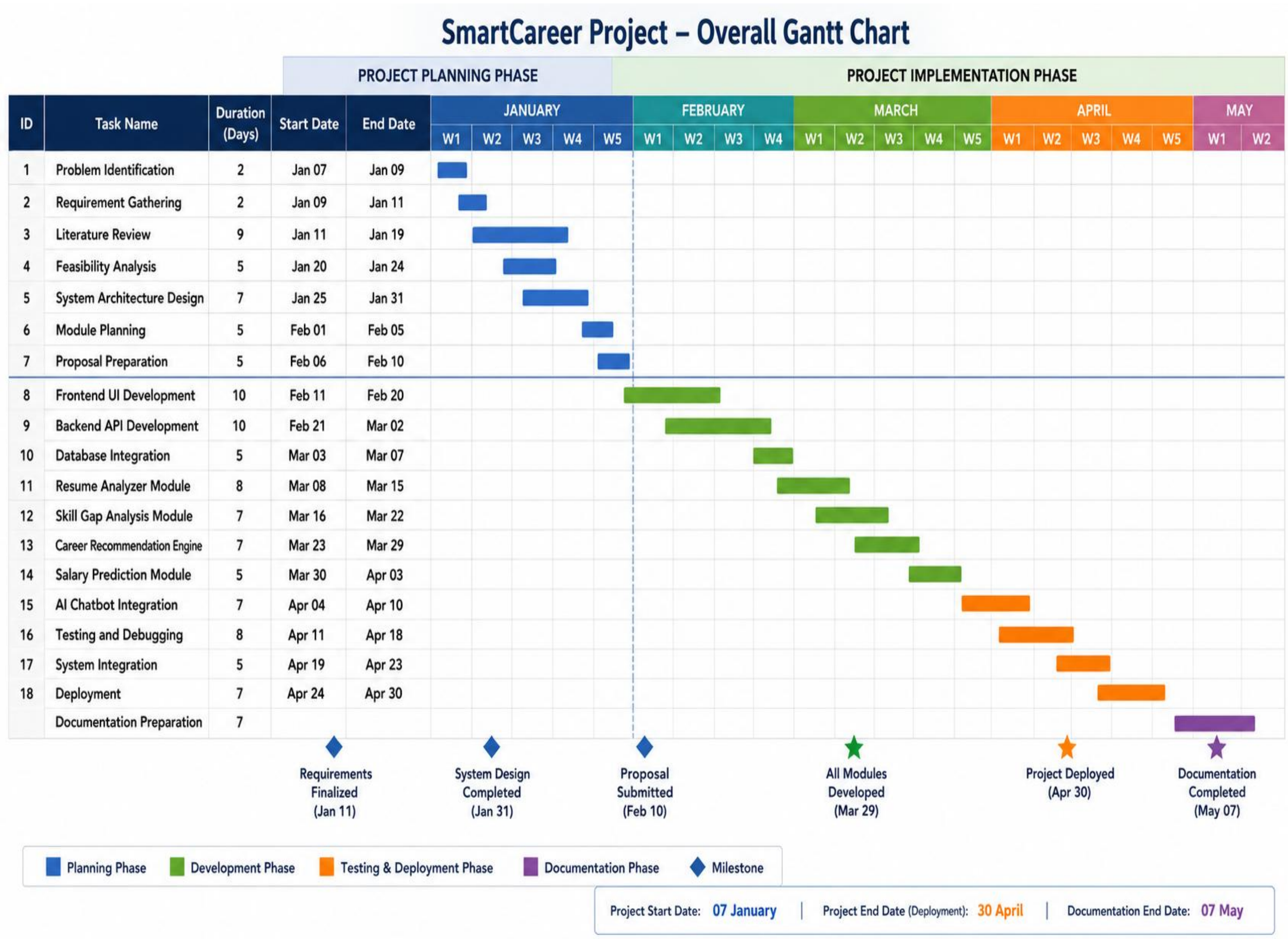

Fig 4.1 Gantt Chart for CareerShala Project

4.2 Risk Analysis

Risk analysis is an essential component of project management that helps identify factors which may affect project success. For the CareerShala project, a **PESTLE analysis** was performed to evaluate external risks and identify mitigation strategies.

PESTLE analysis includes:

- Political Factors
- Economic Factors
- Social Factors
- Technological Factors
- Legal Factors
- Environmental Factors

This analysis helps in proactive planning and improves the reliability and sustainability of the project.

Table 4.2 PESTLE Analysis for CareerShala System

Factor	Impact on Project	Risk Mitigation Strategy
Political	Changes in educational policies	Continuous updates to system data
Economic	Budget limitations	Use of open-source technologies
Social	Lack of user awareness	User-friendly interface and guidance
Technological	Rapid technology changes	Regular software updates
Legal	Privacy and data security issues	Secure authentication and encryption
Environmental	Increased server usage	Efficient cloud resource management

Description

Table 4.2 summarizes the PESTLE analysis conducted for the CareerShala system. The analysis identifies possible external risks and corresponding mitigation strategies. The use of open-source technologies, secure authentication, and regular updates reduces the impact of these risks and improves overall project reliability.

4.3 Project Budget

Budget estimation is necessary to ensure efficient allocation of project resources. The CareerShala project budget includes expenses related to hardware, software, internet usage, hosting services, and documentation.

The following steps were followed while preparing the project budget:

1. List all tasks and required resources
2. Check team and system availability
3. Estimate task duration
4. Use previous experience and available data
5. Set the project budget
6. Track expenses during implementation

Table 4.3 Project Budget Estimation

Sl. No	**Resource**	**Estimated Cost (₹)**
1	Laptop/System	55,000
2	Internet Charges	3,000
3	Cloud Hosting Services	5,000
4	Software Tools & APIs	3,000
5	Database Services	2,500
6	Documentation & Printing	2,000
7	Miscellaneous Expenses	4,500
	Total Estimated Cost	**75,000**

Description

Table 4.3 presents the estimated project budget for the CareerShala system. Since most technologies used in the project are open-source, the overall development cost was reduced significantly. The budget mainly includes hardware, hosting, internet, and documentation expenses required for project completion.

4.4 Summary

This chapter discussed the project management activities involved in the CareerShala system. A structured project timeline was prepared using planning and implementation schedules. Gantt Charts were used to visually represent project tasks and milestones. Risk analysis was conducted using the PESTLE model to identify possible challenges and mitigation strategies. Finally, a project budget was estimated to ensure proper resource allocation and cost management throughout the development lifecycle.

Chapter 5

Analysis and Design

Analysis and design are two closely related phases in system development that ensure a well-structured and efficient solution. The analysis phase focuses on understanding the real-world problem, identifying user needs, and defining system requirements. In contrast, the design phase transforms these requirements into a structured solution by defining system architecture, modules, workflows, and interfaces.

For the CareerShala – One-Stop Personalized Career & Education Advisor Software, analysis involves identifying user expectations such as career guidance, resume analysis, and skill gap identification. The design phase translates these requirements into modules like AI Advisor, Resume Analyzer, Skill Gap Analyzer, Salary Predictor, and Career Sandbox.

5.1 Requirements

The CareerShala system is designed to provide intelligent, personalized career guidance using AI and data-driven insights. The system requirements are categorized into hardware, software, data, management, security, and UI requirements.

A. Functional Requirements

The system must perform the following key functions:

- Allow users to register and log in securely
- Enable users to create and manage their profiles (education, skills, experience)
- Provide a feature to upload resumes (PDF/DOCX)
- Perform AI-based resume analysis including ATS score and keyword matching
- Conduct career assessment quizzes to evaluate user interests
- Generate career recommendations based on user data
- Perform skill gap analysis and suggest improvements
- Predict salary based on role, skills, and experience
- Provide an AI chatbot for career guidance
- Recommend courses and learning paths
- Allow users to download reports and insights

B. Non-Functional Requirements

- **Usability:** System should be easy to use and user-friendly
- **Performance:** Fast response time for real-time recommendations
- **Scalability:** Should support a large number of users
- **Security:** Secure authentication and encrypted data storage
- **Reliability:** High availability with minimal downtime

C. Data Collection Requirements

a) User profile data (education, skills, experience)
b) Resume documents (PDF/DOCX)
c) Job market data and salary datasets
d) Course and certification databases

D. Data Analysis Requirements

a) NLP-based resume parsing
b) Skill matching algorithms
c) Machine learning models for salary prediction
d) Career recommendation logic

E. System Management Requirements

a) Admin dashboard for monitoring system performance
b) User profile management and updates
c) Module-wise tracking of progress

F. Security Requirements

a) User authentication and authorization
b) Secure data storage and encryption
c) Protection against unauthorized access

G. User Interface Requirements

a) Responsive dashboard
b) Interactive charts and visualizations
c) Easy navigation with sidebar modules

d) Clean and modern UI design

Table 5.1 Summarizing Requirements

Category	**Description**
Purpose	Provide personalized career guidance and education recommendations using AI
Behaviour	System analyzes user data and dynamically updates career insights
System Management	Enables tracking of user progress and system performance
Data Analysis	Uses AI/ML for resume analysis, skill gap detection, and predictions
Deployment	Web-based application accessible across devices
Security	Ensures authentication, data privacy, and secure communication

5.2 Use Case Diagram

The use case diagram represents the interaction between users and the CareerShala system. It helps in understanding the functionalities offered by the system and how different actors utilize them.

In this system, the primary actor is the User, who interacts with the application to receive career guidance. A secondary actor (Admin) may be included to manage system operations, data, and analytics.

Actors Involved

User: A student or professional seeking career guidance

Admin: Responsible for managing system data, monitoring analytics, and maintaining the platform

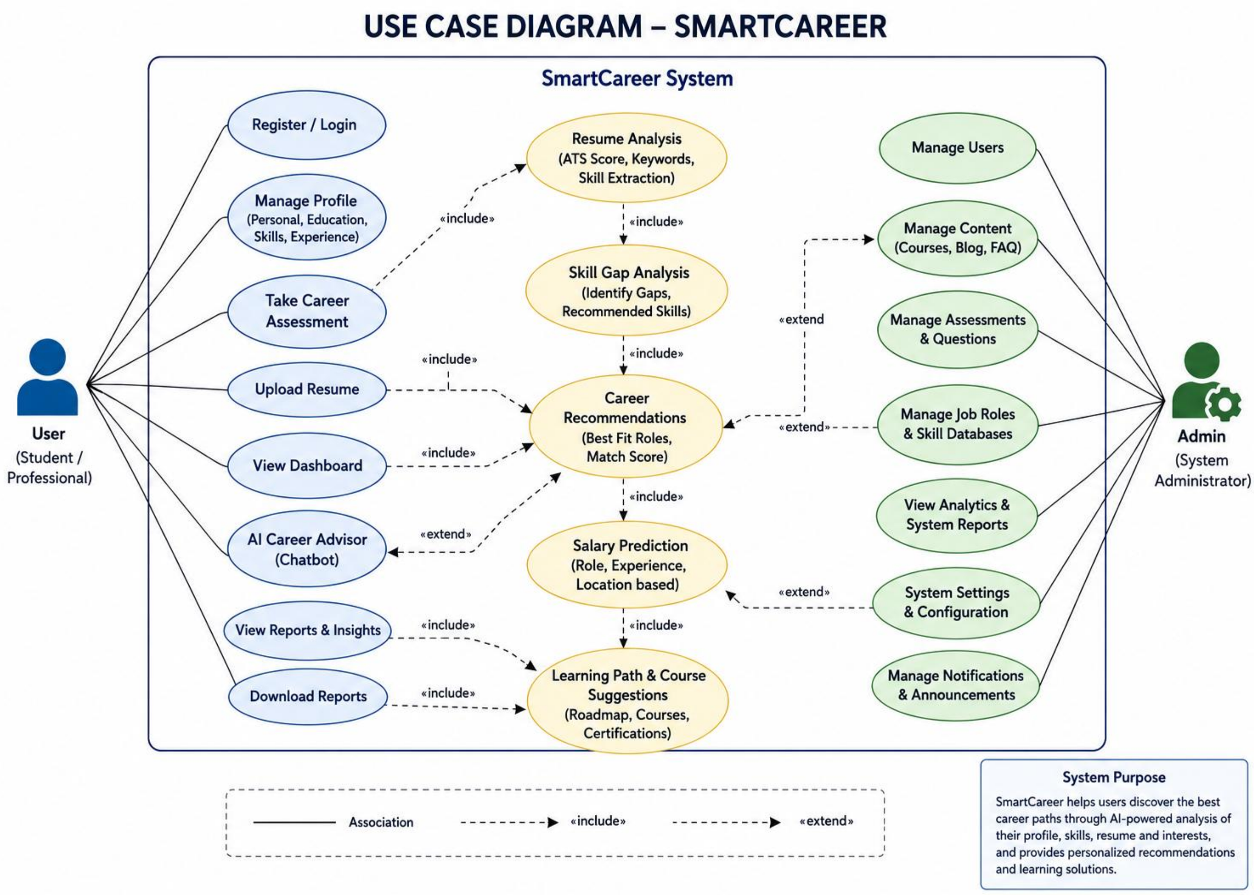

Fig 5.2 : Use Case Diagram

The use case diagram represents the interaction between users and the CareerShala system and provides a clear understanding of how various functionalities are accessed and utilized. In this system, the primary actor is the user, who may be a student or professional seeking career guidance, while a secondary actor, such as an administrator, is responsible for managing system operations, monitoring analytics, and maintaining platform performance. The diagram illustrates how users interact with different modules through a unified interface, enabling seamless access to various services provided by the system.

The use case diagram highlights the relationship between the user and the system functionalities, demonstrating how inputs are provided and processed to generate outputs. The user interacts with multiple features of the system, beginning with authentication, where they can register a new account and log in securely. Once authenticated, users can manage their profiles by entering personal details, adding educational background, skills, and updating their preferences. This information serves as the foundation for further analysis and recommendations.

The system also allows users to upload their resumes in formats such as PDF or DOC, which are then processed using natural language processing techniques to extract relevant information and generate an ATS score. The career assessment module enables users to answer questionnaires that evaluate their interests and aptitude, leading to the generation of career match scores. Based on this analysis, the system performs skill gap evaluation by comparing user skills with industry requirements, identifying missing competencies, and suggesting a roadmap for improvement.

Furthermore, the system provides career recommendations by suggesting suitable job roles and career paths along with match percentages. The salary prediction module estimates potential earnings based on user inputs and presents insights into growth trends and market conditions. The AI career advisor enhances user interaction by offering chatbot-based guidance, answering queries, and providing real-time support. Additionally, the system recommends learning paths by suggesting courses and certifications tailored to the user's needs.

Finally, the system includes a report generation feature that allows users to generate and download detailed reports containing insights, recommendations, and analysis results. Overall, the use case diagram demonstrates how the CareerShala system integrates multiple modules into a cohesive platform, ensuring that users receive comprehensive and personalized career guidance through a seamless and interactive experience.

5.3 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) provides a structured representation of how data moves through the CareerShala system and how it is transformed from raw inputs into meaningful outputs. It explains the internal functioning of the system by illustrating the flow of information between different components, including input processing, AI-based analysis modules, data storage, and output generation. The diagram helps in understanding how user data is handled at each stage and how various modules interact to produce personalized career insights.

The diagram begins with the user, who acts as the primary external entity and provides input data to the system. This input includes profile information, resume or CV, user preferences, assessment responses, and details about skills and experience. At the highest level, known as the Level 0 DFD or context diagram, the entire system is represented as a single process called the CareerShala System. In this level, all user inputs are processed collectively, and the system generates outputs such as career recommendations, skill gap reports, salary insights, learning paths, and AI advisor responses. This level provides a simplified overview of the system without detailing internal processes.

At the next level, known as the Level 1 DFD, the system is broken down into multiple detailed processes that explain how data is handled internally. The first stage is input processing, where the system validates user inputs, parses resume data using text processing techniques, extracts key information such as skills and experience, and stores the initial data in the user database. This stage ensures that all incoming data is structured and ready for further analysis.

Following this, the processed data is passed to the AI processing modules, which form the core of the CareerShala system. These modules include the resume analyzer, which calculates the ATS score and extracts relevant keywords; the skill gap analyzer, which identifies missing skills and suggests improvements; the career recommendation engine, which matches user profiles with suitable job roles; the salary predictor, which estimates expected earnings based on user inputs; and the AI advisor, which provides real-time guidance using natural language processing. Each of these modules processes the data independently while contributing to a unified output.

The next stage involves data storage, where different types of data are stored in structured databases. The user database stores profile information, skills, and preferences, while the processing database maintains extracted data, analysis results, and predictions. Additionally, an analytics database stores historical records, user interactions, and recommendation data, enabling the system to track user progress and improve performance over time. This layered storage approach ensures efficient data management and retrieval.

After processing and storage, the system moves to the output generation stage, where meaningful insights are created and presented to the user. This includes generating dashboard insights, displaying career recommendations, creating skill gap reports, providing salary predictions, suggesting learning paths, and producing downloadable reports. The outputs are designed to be clear, interactive, and easy to understand, ensuring a user-friendly experience.

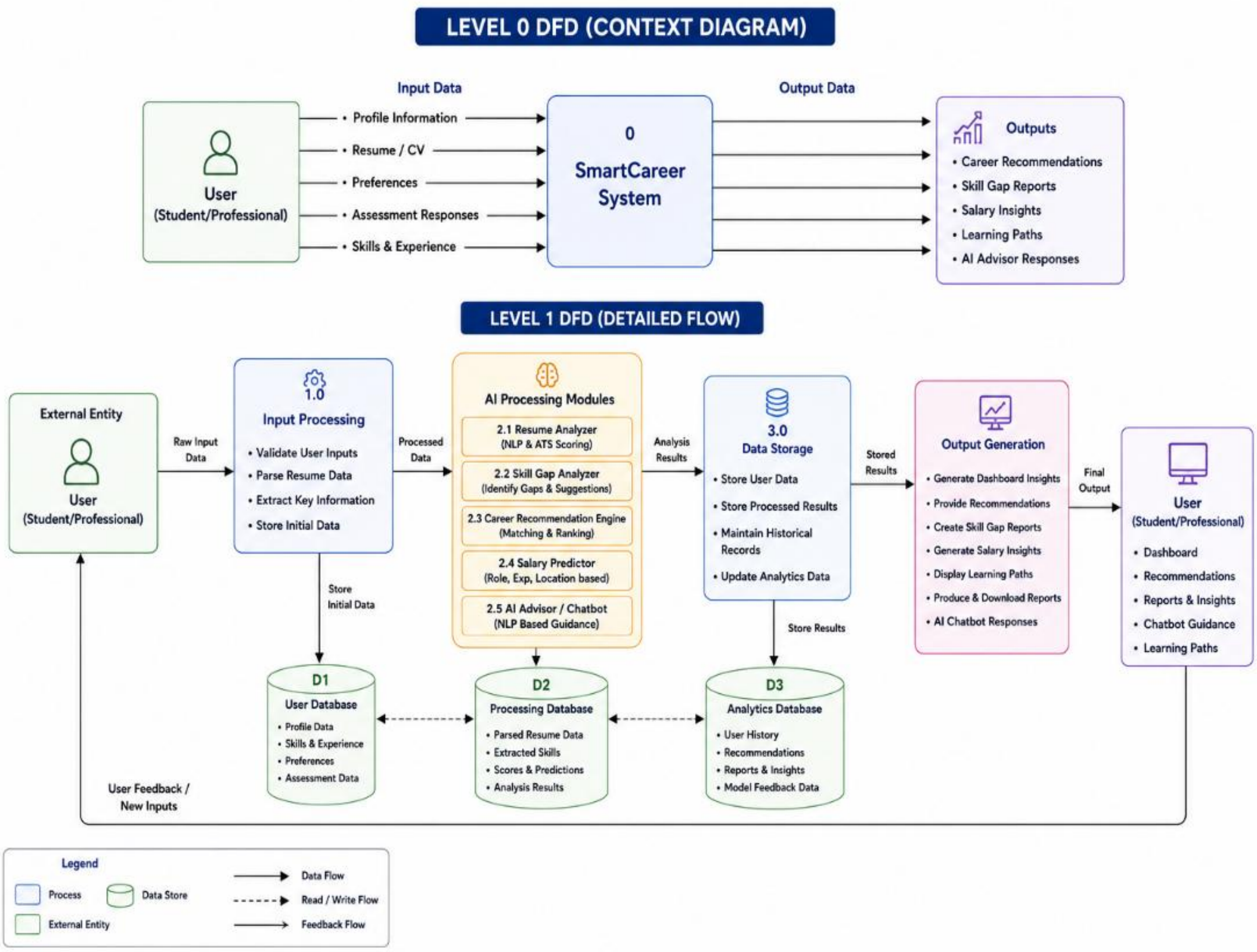

Fig 5.3 :Level 0 DFD

DFD Image Description

The DFD image clearly illustrates the complete flow of data within the CareerShala system, starting from user input and progressing through multiple processing layers to generate final outputs. It shows how raw input data is first validated and structured, then analyzed using AI modules, and finally stored in different databases for efficient management. The diagram highlights the interaction between components such as input processing, AI modules, data storage, and output generation, demonstrating a well-organized system architecture. It also emphasizes the continuous feedback loop, where user interactions and new inputs are used to update the system and improve future recommendations. Overall, the diagram effectively represents how the CareerShala platform transforms user data into personalized career insights in a systematic and efficient manner.

5.4 System Architecture / Block Diagram

The system architecture of the CareerShala platform defines the overall structure and interaction between different components of the system. It provides a high-level view of how data flows from the user interface through various processing modules and finally produces meaningful outputs. The architecture is designed in a modular and scalable manner so that different functionalities such as resume analysis, skill gap detection, career recommendation, salary prediction, and AI advisory services can work independently while still being integrated into a unified system.

At the input level, the system accepts user data such as personal profile information, educational background, technical and soft skills, resume documents, and responses from career assessment questionnaires. This data acts as the primary input for the system and is essential for generating personalized career insights. The input data is first passed through the input processing layer, where validation, cleaning, and preprocessing operations are performed to ensure data accuracy and consistency. Resume files are processed using Natural Language Processing (NLP) techniques to extract relevant information such as skills, experience, and qualifications.

The core of the system consists of multiple AI-driven processing modules. These modules include the Resume Analyzer, Skill Gap Analyzer, Career Recommendation Engine, Salary Predictor, and AI Career Advisor. Each module performs a specific function. The resume analyzer extracts structured data from resumes, while the skill gap analyzer compares user skills with industry requirements to identify missing competencies. The career recommendation engine suggests suitable job roles based on user profiles, and the salary predictor estimates potential earnings using machine learning models. The AI advisor module provides real-time guidance through chatbot interaction, helping users clarify doubts and explore career options.

The processed data is then stored in the data storage layer, which includes multiple databases such as user database, processing database, and analytics database. The user database stores personal information, preferences, and historical data. The processing database contains extracted features, analysis results, and model outputs, while the analytics database maintains insights, reports, and user interaction history. This layered storage approach ensures efficient data management, quick retrieval, and system scalability.

After processing and storage, the output generation layer presents the results to the user through an interactive dashboard. The system generates outputs such as career match scores, skill gap reports, recommended career paths, salary insights, learning roadmaps, and downloadable reports. Visual elements like graphs, charts, and heatmaps are used to improve user understanding and engagement. The frontend interface communicates with the backend services through APIs, ensuring smooth and real-time data exchange.

Overall, the CareerShala system architecture follows a layered approach consisting of input, processing, storage, and output layers. This design ensures modularity, scalability, and efficient performance. It allows the system to handle multiple functionalities simultaneously while maintaining accuracy and responsiveness. The architecture also supports future enhancements such as integration with real-time job portals, advanced AI models, and cloud-based deployment, making it a robust and adaptable solution for personalized career guidance.

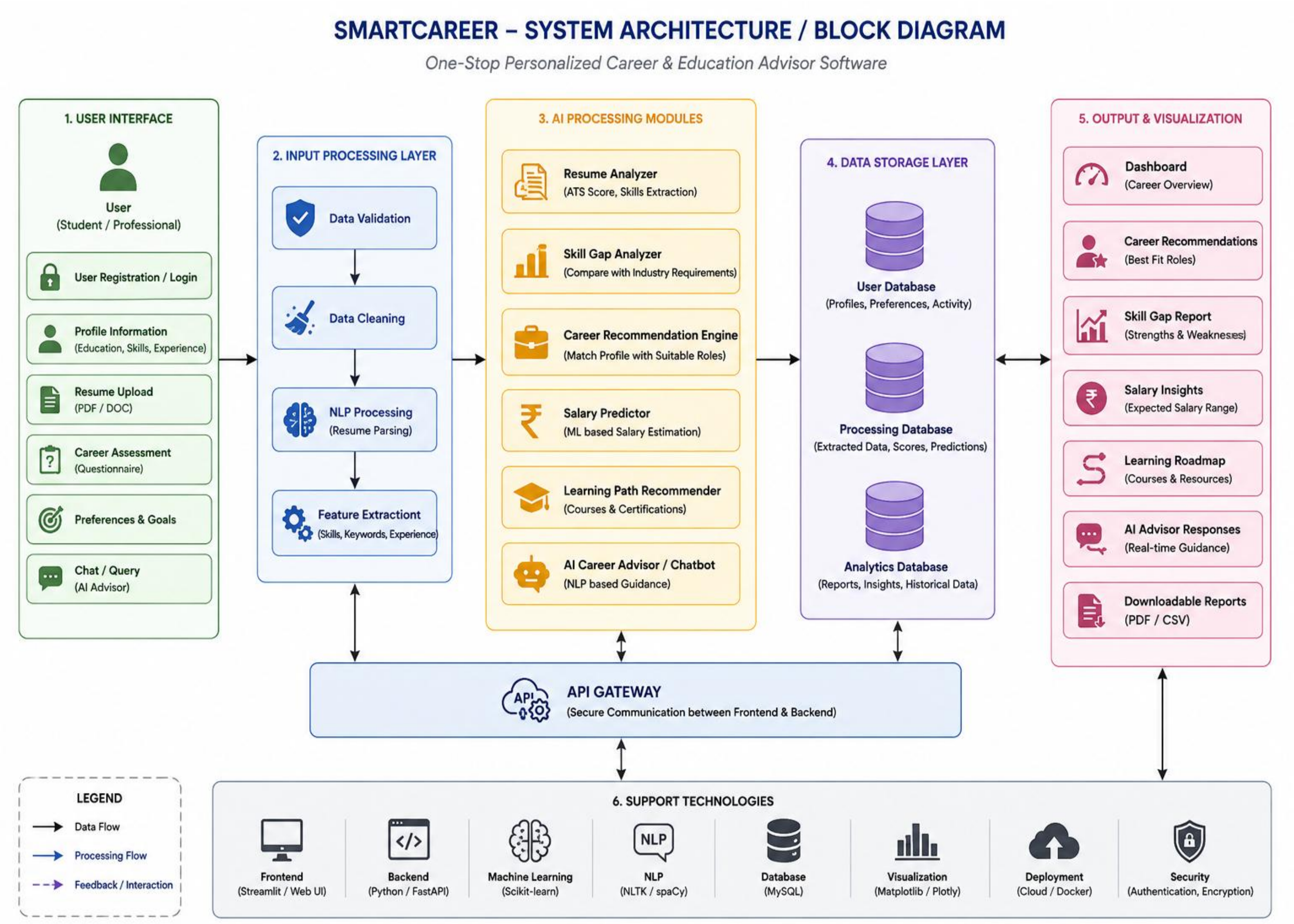

Figure 5.4 System Architecture / Block Diagram (Description)

Figure 5.4 illustrates the system architecture of the CareerShala platform, showing the flow of data from user input through various AI processing modules to the final output generation stage. It highlights how different components such as input processing, AI modules, data storage, and output visualization interact to deliver personalized career recommendations and insights.

5.5 Database Design (ER Diagram & Schema)

The database is designed to store structured data efficiently and support fast retrieval of information.

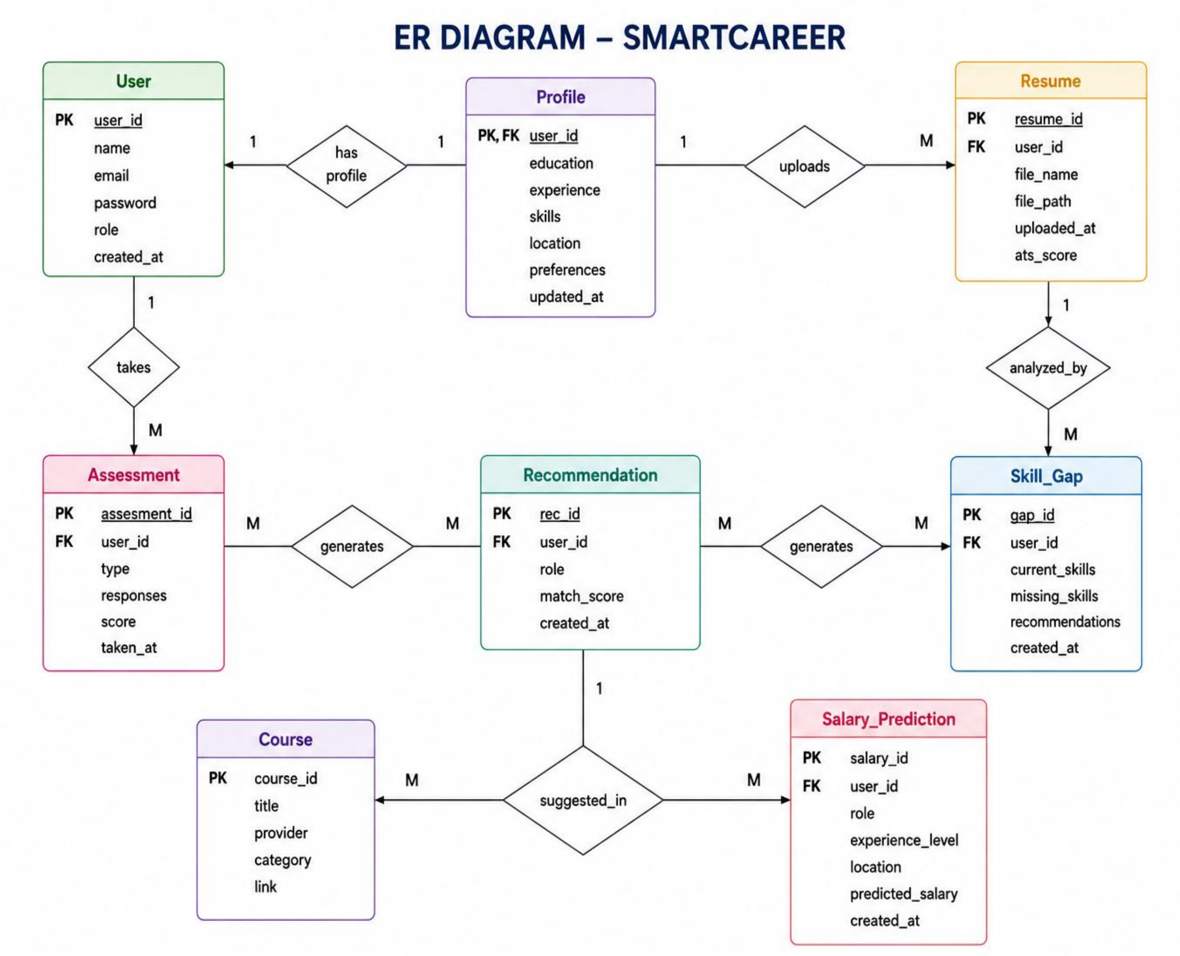

Fig 5.5 : ER Diagram CareerShala

Description

This ER diagram illustrates the structured database design of the CareerShala system, highlighting the core entities such as User, Profile, Resume, Assessment, Recommendation, Skill Gap, Course, and Salary Prediction. Each entity contains relevant attributes and is connected through defined relationships, representing how user information is stored, managed, and processed within the system. The diagram clearly shows how a user creates a profile, uploads resumes, and participates in assessments, forming the foundation for further analysis.

Additionally, the diagram demonstrates how processed data flows between modules to generate meaningful outputs such as career recommendations, skill gap analysis, and salary predictions. Relationships like "generates," "uploads," and "analyzed_by" indicate how different components interact to provide personalized insights. Overall, the ER diagram ensures efficient data organization, supports scalability, and enables seamless integration of AI-driven features within the CareerShala platform.

Relationships

- One user has one profile
- One user can have multiple assessments
- One user gets multiple recommendations
- Skills are linked to profiles

Database Description

The database ensures data consistency and supports efficient querying. It is designed to handle large datasets and maintain relationships between different entities

5.6 User Interface Design

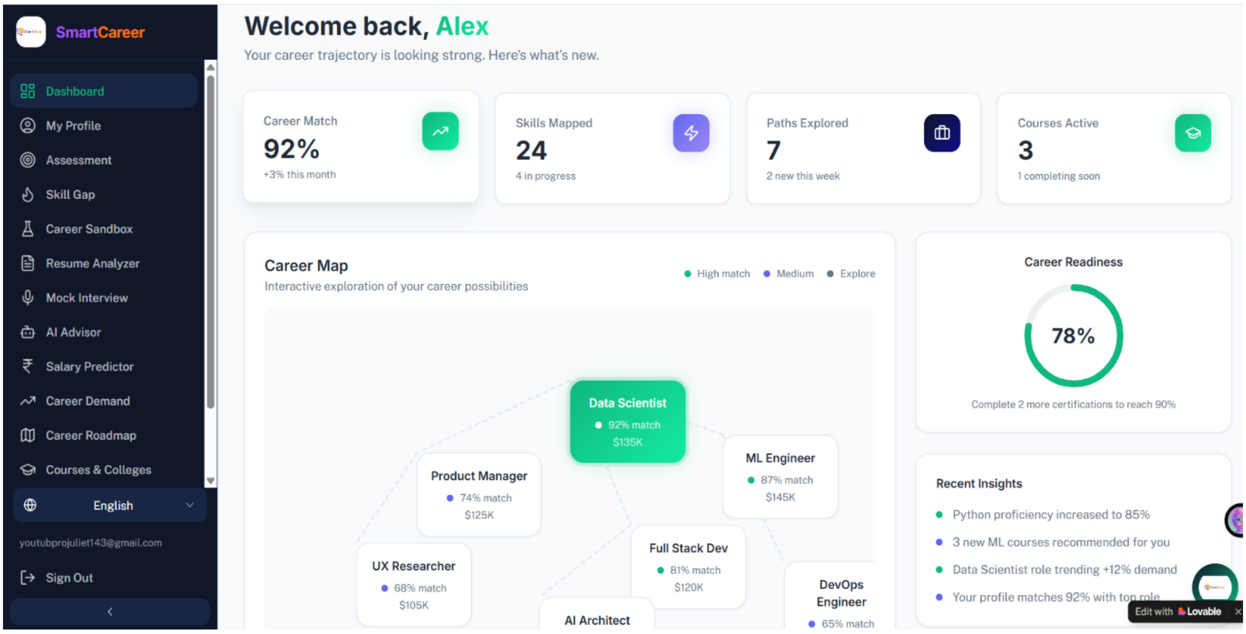

Figure 5.6 User Interface Design of CareerShala System

Description

The user interface of the CareerShala system is designed to provide a clean, intuitive, and user-friendly experience for students and professionals seeking career guidance. The dashboard layout presents key insights such as profile strength, skill analysis, and recommended career actions in a structured and visually appealing manner, enabling users to quickly understand their progress and areas of improvement.

The interface includes multiple sections such as resume analysis, skill gap visualization, and recommended next steps, all organized using cards, charts, and progress indicators. The navigation panel on the left allows users to easily switch between modules like profile management, resume analyzer, and career recommendations, ensuring smooth accessibility and seamless interaction across the platform.

Key UI Features

- **Dashboard Overview:** Displays profile strength, skills identified, and assessment status
- **Resume Analysis Section:** Shows ATS score, strengths, and improvement suggestions
- **Skill Gap Visualization:** Uses progress bars to highlight missing and strong skills
- **Recommendations Panel:** Suggests learning paths and career improvement steps
- **Navigation Menu:** Provides easy access to all system modules
- **Responsive Design:** Works efficiently across desktop and mobile devices

Suitability for the Project

The user interface is specifically designed to simplify complex career data into easy-to-understand visuals. By using interactive components such as charts, progress bars, and cards, the system ensures better user engagement and improves decision-making for career planning. This design approach enhances usability and makes the platform accessible even for non-technical users.

5.7 Technology Stack and Standards Followed

5.7.1 Technology Stack

The CareerShala system uses modern technologies to ensure performance and scalability.

- Frontend: React.js, HTML, CSS, Tailwind CSS
- Backend: Python, Django
- Database: MongoDB / PostgreSql
- AI/ML: Python, Scikit-learn, NLP libraries
- Visualization: Chart.js, Recharts
- Deployment: Vercel / Netlify

5.7.2 Standards Followed

- Modular coding for maintainability
- Clean and readable code practices
- RESTful API design
- Error handling and logging
- Secure authentication mechanisms
- Scalable architecture design

Standards Description

These standards ensure that the system remains robust, easy to maintain, and scalable for future enhancements.

CHAPTER 6
IMPLEMENTATION AND SIMULATION

6.1 Hardware / Software Requirements

The CareerShala system is implemented as a web-based application, making it accessible to users through standard computing devices such as laptops, desktops, and smartphones. The hardware requirements are relatively minimal, ensuring that the system can be used by a wide range of users without the need for specialized equipment. A system with at least 8 GB of RAM, a dual-core processor, and stable internet connectivity is sufficient to ensure smooth operation. Since the application is designed to run in a browser environment, users only require an updated web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge to access the platform.

From a software perspective, the system relies on a combination of programming languages, libraries, and frameworks that support efficient development and execution. Python is used as the core programming language due to its flexibility and strong support for machine learning and natural language processing. The application leverages various libraries for data handling, analysis, and visualization, which enables the system to process user inputs effectively and generate meaningful insights. The use of modern software tools ensures that the system remains scalable and adaptable to future technological advancements.

6.2 Development Tools and Frameworks Used

The development of the CareerShala system is carried out using a combination of advanced tools and frameworks that facilitate rapid development and efficient execution. The frontend interface is developed using Streamlit, which allows for the quick creation of interactive web applications with minimal coding effort. Streamlit provides built-in components for displaying charts, forms, and interactive elements, making it suitable for presenting career insights and recommendations in a visually appealing manner.

On the backend, Python serves as the primary programming language, supporting the integration of various libraries for data processing and machine learning. Natural Language Processing is implemented using NLTK, regular expressions, and text processing techniques to extract relevant information from resumes. Machine learning models are developed using Scikit-learn, which enables tasks such as classification and prediction, including salary estimation and career recommendations. Visualization libraries such as Matplotlib and WordCloud are used to present data in graphical formats, enhancing user understanding. The entire development process is managed using PyCharm IDE, which provides a comprehensive environment for coding, debugging, and testing.

6.3 Implementation Details (Modules and Functions)

The CareerShala system is implemented using a modular approach, where each module is responsible for a specific functionality while interacting with other components to form a complete system. The user profile module is designed to collect and manage user information, including educational background, skills, and career preferences. This information serves as the foundation for all subsequent analysis and recommendations provided by the system.

The resume analyzer module plays a crucial role in extracting meaningful information from user-uploaded resumes. Using natural language processing techniques, the system identifies key skills, experience, and keywords, which are then used to calculate an ATS score. This score indicates how well the resume aligns with industry standards and provides users with suggestions for improvement. The career assessment module evaluates user interests and aptitudes through a structured questionnaire, helping to identify suitable career paths.

The skill gap analysis module compares the user’s current skills with industry requirements and identifies areas that need improvement. This module provides a visual representation of skill gaps, enabling users to understand their strengths and weaknesses effectively. The career recommendation module uses machine learning algorithms and rule-based logic to suggest job roles that match the user’s profile. Additionally, the salary prediction module estimates expected salaries based on factors such as role, experience, and skill set, providing users with insights into their earning potential.

The AI career advisor module enhances user interaction by providing real-time guidance through a chatbot interface. It answers user queries, offers suggestions, and helps users navigate through the platform. All these modules are integrated into a unified system that ensures smooth data flow and consistent output generation.

6.4 Simulation and Testing Environment

The CareerShala system is tested in a simulated environment to ensure that all components function correctly under different conditions. The application is executed locally using a Streamlit server, allowing developers to test the system in real time. Sample datasets, including resumes and job descriptions, are used to simulate real-world scenarios and validate the system's performance.

The testing process involves evaluating each module individually as well as in an integrated environment. Various test cases are executed to verify the correctness of data processing, accuracy of recommendations, and responsiveness of the user interface. Edge cases, such as incomplete inputs and incorrect data formats, are also tested to ensure that the system handles errors gracefully. The simulation environment plays a vital role in identifying and resolving issues before deployment, ensuring that the final system is reliable and efficient.

CHAPTER 7
EVALUATION AND RESULTS

7.1 Test Plan and Test Cases

Testing is a crucial phase in the development of the CareerShala system, as it ensures that all modules function correctly and meet the specified user requirements. The testing process helps in identifying system errors, validating functionalities, and improving the overall reliability and performance of the application. The CareerShala system is tested using multiple testing approaches, including functional testing, integration testing, user interface testing, and performance testing. These testing methods ensure that all components such as frontend interface, backend processing, AI modules, and database integration work efficiently and cohesively.

The system is evaluated by testing core functionalities such as user authentication, resume analysis, skill gap detection, career recommendation generation, salary prediction, and AI chatbot interaction. The testing process verifies the correctness of data processing, ensures smooth communication between modules, and confirms that outputs are accurate and meaningful.

7.1.1 Test Plan

The test plan outlines the objectives, testing strategies, modules tested, and expected outcomes of the CareerShala system. It ensures that all aspects of the system are thoroughly validated before deployment.

Objectives of Testing

The primary objective of testing is to verify the functionality of each module, ensure accurate AI-based analysis, validate database operations, and confirm smooth user interaction through the dashboard interface. The system is also tested to identify and resolve errors, improve performance, and ensure reliability under different conditions.

Table 7.1 Test Plan

Testing Type	**Purpose**
Functional Testing	Verify system functionalities
Integration Testing	Validate communication between modules

UI Testing	Check user interface responsiveness
Database Testing	Verify data storage and retrieval
Performance Testing	Analyze system efficiency and speed

7.1.2 Test Cases

Test cases are designed to validate individual modules of the CareerShala system and ensure that each functionality performs as expected. These test cases cover different user interactions and system processes.

Table 7.2 Test Cases

Test Case ID	Module	Test Description	Expected Result	Status
TC01	Login Module	Verify user login functionality	Successful login	Pass
TC02	Profile Module	Add and update user profile data	Profile updated successfully	Pass
TC03	Resume Analyzer	Upload resume and extract skills	Skills extracted and ATS score generated	Pass
TC04	Skill Gap Analyzer	Compare skills with job requirements	Missing skills identified	Pass
TC05	Career Recommendation	Generate job role suggestions	Relevant roles recommended	Pass
TC06	Salary Predictor	Predict salary based on user input	Salary estimate displayed	Pass
TC07	AI Advisor	Ask career-related queries	Relevant responses generated	Pass
TC08	Report Generation	Generate downloadable report	Report generated successfully	Pass

The testing workflow ensures that all modules of the CareerShala system are validated, and the application performs efficiently under different scenarios.

7.2 Test Results and Analysis

The CareerShala system was successfully executed in the testing environment, and all major modules performed effectively. The frontend dashboard displayed user information, career insights, and visual analytics accurately. The backend system processed user inputs efficiently and ensured smooth communication with AI modules and databases.

The resume analyzer successfully extracted relevant skills and generated ATS scores, while the skill gap analyzer accurately identified missing competencies and suggested improvement strategies. The career recommendation engine provided meaningful job role suggestions based on user profiles, and the salary predictor generated realistic salary estimates aligned with market trends. The AI advisor module responded effectively to user queries, enhancing user interaction.

The database system efficiently stored and retrieved user data, ensuring consistency and reliability. Integration testing confirmed that all modules worked together seamlessly without major issues.

Table 7.3 Test Results

Module Tested	**Result Obtained**	**Status**
Frontend Dashboard	Responsive UI displayed	Successful
Backend Processing	Data processed correctly	Successful
Database Integration	Data stored and retrieved successfully	Successful
Resume Analyzer	Skills extracted and ATS score generated	Successful
Skill Gap Analyzer	Missing skills identified	Successful
Career Recommendation	Relevant roles suggested	Successful
Salary Predictor	Salary estimates generated	Successful
AI Advisor	Queries answered correctly	Successful

7.3 Performance Evaluation

Performance evaluation is conducted to measure the efficiency, speed, accuracy, and stability of the CareerShala system. The system is analyzed based on parameters such as frontend response time, backend processing efficiency, database query performance, and AI model accuracy.

The frontend dashboard demonstrated fast loading and smooth navigation, allowing users to interact with different modules without delays. Backend processing was efficient, handling user requests and generating responses quickly. Database operations were stable, ensuring fast storage and retrieval of data. The AI modules provided accurate recommendations and predictions, contributing to the overall effectiveness of the system.

Performance Parameter	**Observation**
Frontend Response Time	Fast
Backend Processing	Efficient
Database Query Execution	Stable
Dashboard Visualization	Responsive
AI Recommendation Accuracy	High

Table 7.4 Performance Evaluation

The system shows strong performance across all parameters, making it suitable for real-time career guidance applications.

7.4 Discussion of Results / Insights

The implementation and testing results demonstrate that the CareerShala system successfully integrates multiple technologies into a unified platform for career guidance. The frontend interface provides a user-friendly experience, while the backend ensures efficient data processing and communication between modules. The AI-based modules enhance the system's ability to provide personalized insights and recommendations.

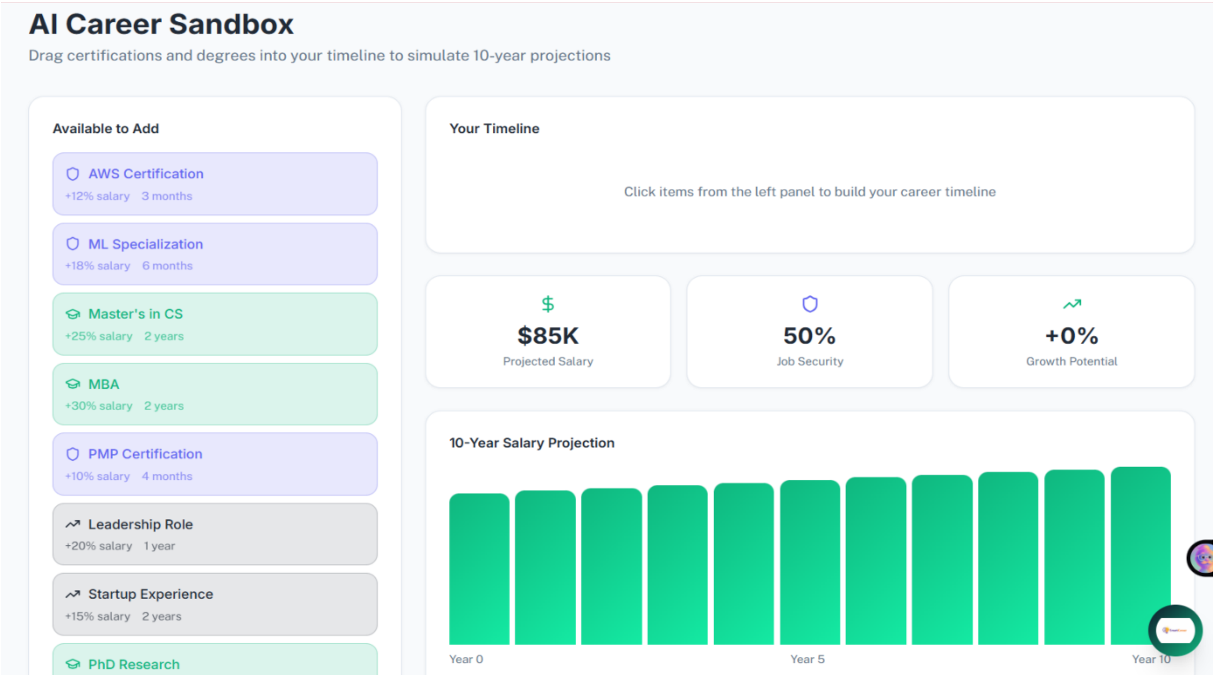

Fig 7.4.1 : CareerShala AI career Sandbox

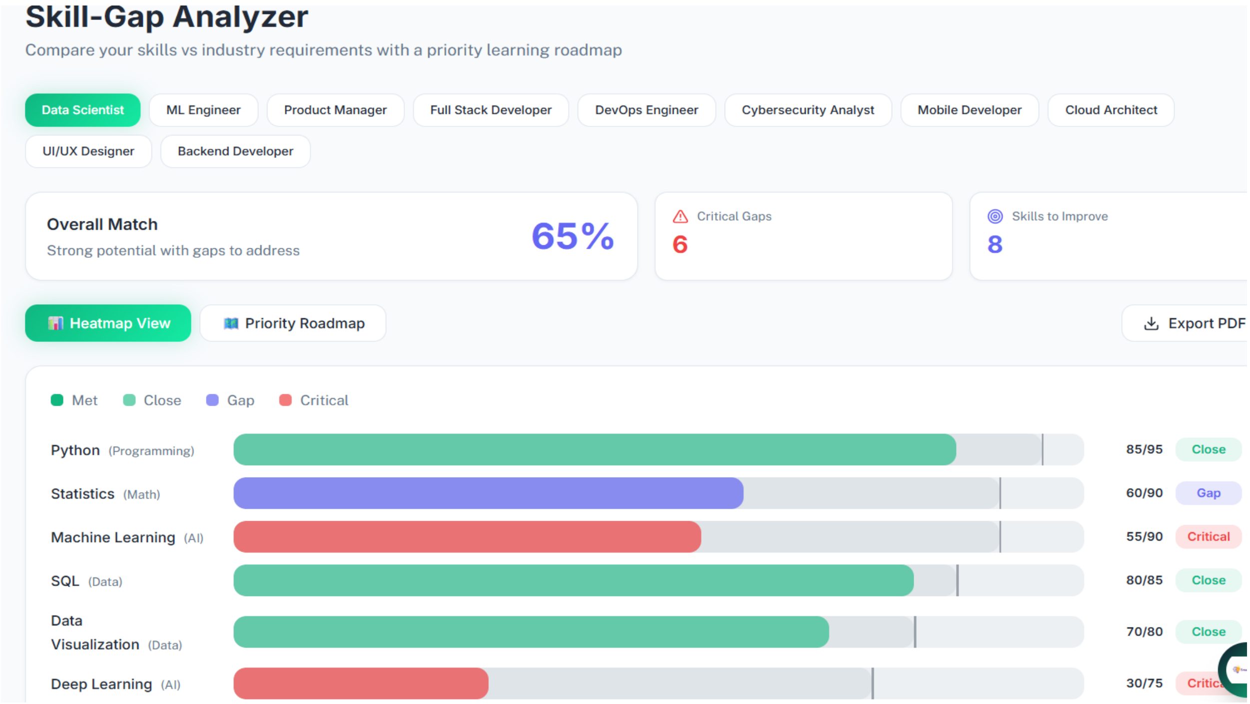

Fig 7.4.2 : CareerShala Skill-Gap Analyer

The results indicate that the system effectively bridges the gap between user skills and industry requirements by identifying skill gaps and suggesting appropriate learning paths. The integration of salary prediction and career recommendation modules adds further value by helping users understand their potential and make informed decisions.

CHAPTER 8
SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

The proposed project titled *"CareerShala: One-Stop Personalized Career & Education Advisor Software"* is developed to provide intelligent career guidance using Artificial Intelligence, Machine Learning, and data analytics techniques. While the technical implementation focuses on building an efficient and user-friendly system, it is equally important to evaluate the broader impact of the system in terms of social, legal, ethical, sustainability, and safety aspects. These considerations ensure that the system is not only technically sound but also responsible, secure, and beneficial to users and society as a whole.

8.1 Social Aspects

The CareerShala system has a significant positive impact on society by improving access to career guidance for students and professionals. Traditionally, career counseling services are limited, expensive, and not easily accessible to everyone. This system addresses these challenges by providing a digital platform where users can receive personalized career recommendations, skill analysis, and learning pathways based on their profile and preferences. As a result, users from different backgrounds can benefit from advanced career guidance without geographical or financial barriers.

The system also promotes digital literacy and awareness of emerging technologies by integrating AI-based solutions into career planning. It helps users understand current industry trends, in-demand skills, and job opportunities, enabling them to make informed decisions about their future. Additionally, the platform supports continuous learning by recommending courses and certifications, which contributes to skill development and employability.

However, there are certain challenges associated with social aspects. Users may become overly dependent on system-generated recommendations and may not explore alternative career options independently. Inexperienced users might misinterpret results or rely solely on the system without considering personal interests or external advice. Therefore, the CareerShala system is designed to act as a guidance tool rather than a replacement for human decision-making, encouraging users to use the recommendations as support rather than final decisions.

8.2 Legal Aspects

Legal aspects play a crucial role in ensuring that the CareerShala system complies with data protection laws, privacy regulations, and software licensing standards. The system collects and processes sensitive user information such as resumes, personal details, and career preferences, making data privacy and security essential components of the design. Proper mechanisms are implemented to ensure that user data is securely stored and accessed only by authorized individuals.

The application is developed using open-source technologies and frameworks, ensuring compliance with their respective licenses. Standard practices such as user authentication, secure data storage, and controlled access mechanisms are implemented to prevent unauthorized usage and protect user information. The system also ensures that user data is not shared with third parties without consent.

If the system is expanded in the future as a large-scale commercial platform, it would need to comply with international and national data protection laws such as the General Data Protection Regulation (GDPR) and India's Digital Personal Data Protection Act (DPDPA). Legal considerations would also include user consent management, data encryption, secure API communication, and compliance with cybersecurity standards. These measures ensure that the system operates within legal boundaries and maintains user trust.

8.3 Ethical Aspects

Ethical considerations are essential in the development of AI-based systems like CareerShala, as they directly influence user decisions and career outcomes. The system provides career recommendations, salary predictions, and skill analysis based on data-driven algorithms. Therefore, it is important to ensure that the recommendations are fair, unbiased, and transparent.

The CareerShala system is designed to provide guidance rather than make decisions on behalf of the user. It does not guarantee job placement or career success, and users are encouraged to validate the recommendations independently. Transparency is maintained by clearly presenting how results are generated and what factors influence the recommendations.

Another important ethical aspect is the prevention of bias in AI models. The system aims to avoid discrimination based on factors such as gender, location, or educational background. Ethical engineering practices such as fairness, accountability, and responsible data usage are followed throughout development. Additionally, user data is handled with strict confidentiality, ensuring that personal information is not misused or manipulated.

Overall, the system promotes responsible AI usage by supporting users in decision-making while respecting human judgment and ethical standards.

8.4 Sustainability Aspects

Sustainability is an important consideration in modern software development, and the CareerShala system is designed to support environmentally friendly and resource-efficient practices. By providing digital career guidance, the system reduces the need for physical documentation, printed reports, and in-person counseling sessions. This contributes to reducing paper usage and supports eco-friendly operations.

The use of lightweight web technologies and modular architecture ensures efficient resource utilization and reduces computational overhead. The system is designed to be scalable, allowing future enhancements without requiring complete redevelopment. This improves long-term sustainability and reduces development costs.

Additionally, the platform encourages continuous learning and skill development, contributing to sustainable career growth for users. By helping individuals adapt to changing industry demands, the system supports long-term employability and professional development. Efficient database management and optimized processing further contribute to reducing energy consumption and improving system performance.

8.5 Safety Aspects

Safety aspects focus on ensuring secure system operation, protecting user data, and maintaining reliable performance. The CareerShala system incorporates authentication mechanisms to ensure that only authorized users can access the platform. Sensitive user information such as resumes and personal data is stored securely, preventing unauthorized access and data breaches.

The system is thoroughly tested to ensure stable execution across all modules, including frontend interface, backend processing, AI models, and database integration. Error handling mechanisms are implemented to manage unexpected inputs and prevent system failures. This ensures that the application remains reliable even under different usage conditions.
Since the system provides guidance rather than executing real-world transactions, it minimizes risks associated with financial or career decisions. The AI recommendations are designed to assist users rather than enforce actions, ensuring that users retain full control over their decisions. Secure communication between system components further enhances safety and reliability.

The modular design also allows developers to identify and fix issues efficiently, improving maintainability and system stability over time. These safety measures ensure that the CareerShala platform remains secure, reliable, and user-friendly.

Conclusion

The CareerShala system successfully incorporates social, legal, ethical, sustainability, and safety considerations into its design and implementation. The platform provides accessible and personalized career guidance while ensuring secure data handling and responsible AI usage. It promotes digital transformation, supports sustainable practices, and enhances user decision-making through intelligent insights.
The project demonstrates how modern AI-based systems can be developed in a socially beneficial, legally compliant, ethically responsible, environmentally sustainable, and technically secure manner. With future enhancements, the CareerShala platform has the potential to become a comprehensive and widely adopted solution for career guidance and professional development.

CHAPTER 9
CONCLUSION AND FUTURE WORK

Conclusion

The CareerShala system successfully demonstrates how artificial intelligence, data analytics, and modern web technologies can be combined to create an intelligent and user-centric career guidance platform. The system integrates multiple modules such as resume analysis, skill gap detection, career recommendation engine, salary prediction, and an AI-powered advisor into a single unified application. This integration enables users to receive personalized insights based on their skills, interests, education, and career goals, thereby supporting informed decision-making.

One of the major achievements of this project is its ability to bridge the gap between user capabilities and industry expectations. Many students and professionals struggle to identify suitable career paths due to a lack of proper guidance and awareness of market trends. The CareerShala system addresses this problem by analyzing user data and mapping it with real-world job requirements, helping users understand where they stand and what improvements are needed. The inclusion of visualization features such as dashboards, career maps, and skill heatmaps further enhances user understanding and engagement.

The system is designed with scalability and flexibility in mind, allowing new modules and features to be integrated without major changes to the existing architecture. The use of AI models and data-driven approaches ensures that recommendations are dynamic and adaptable to evolving industry trends. Additionally, the user-friendly interface and modular backend structure make the system efficient, maintainable, and easy to use.

Overall, the CareerShala platform proves to be a practical and effective solution for personalized career planning. It not only assists users in identifying suitable career paths but also provides actionable insights through learning recommendations and skill development strategies. The project highlights the potential of intelligent systems in transforming traditional career guidance into a smart, automated, and accessible service.

Future Work

Although the CareerShala system provides a strong foundation for AI-driven career guidance, there are several opportunities for further enhancement and expansion. One of the most important future improvements is the integration of **real-time job market data**. By connecting the system with job portals such as LinkedIn, Naukri, Indeed, or company APIs, the platform can provide users with live job listings that match their skills and preferences. This would enable users not only to explore career paths but also to directly apply for relevant opportunities, making the system more practical and industry-oriented.

Another key enhancement is the use of **advanced AI and deep learning models** for improved accuracy in predictions and recommendations. Current models can be upgraded using techniques such as transformer-based NLP models, recommendation systems, and predictive analytics to provide more precise career matching, better resume analysis, and personalized learning paths. Incorporating adaptive learning algorithms can further improve the system by continuously learning from user interactions and feedback.

The system can also be extended by adding **multilingual support**, making it accessible to a wider audience across different regions. This would allow users to interact with the platform in their preferred language, thereby improving usability and inclusivity. Additionally, integrating **voice-based interaction** and chatbot enhancements can make the AI advisor more interactive and user-friendly.

Another promising direction is the development of a **real-time career tracking and progress monitoring system**, where users can track their skill improvement, completed courses, certifications, and career milestones over time. Gamification elements such as badges, progress scores, and achievements can be introduced to increase user engagement and motivation.
The platform can also be enhanced with **integration of online learning platforms** such as Coursera, Udemy, and edX, enabling users to directly enroll in recommended courses. Furthermore, incorporating **mock interview simulations with AI feedback**, real-time coding practice environments, and portfolio-building tools can make the system more comprehensive. From an industry perspective, adding **recruiter and company modules** would allow organizations to identify suitable candidates based on skill matching, creating a bridge between job seekers and employers. This would transform the system into a complete ecosystem for career development and recruitment.

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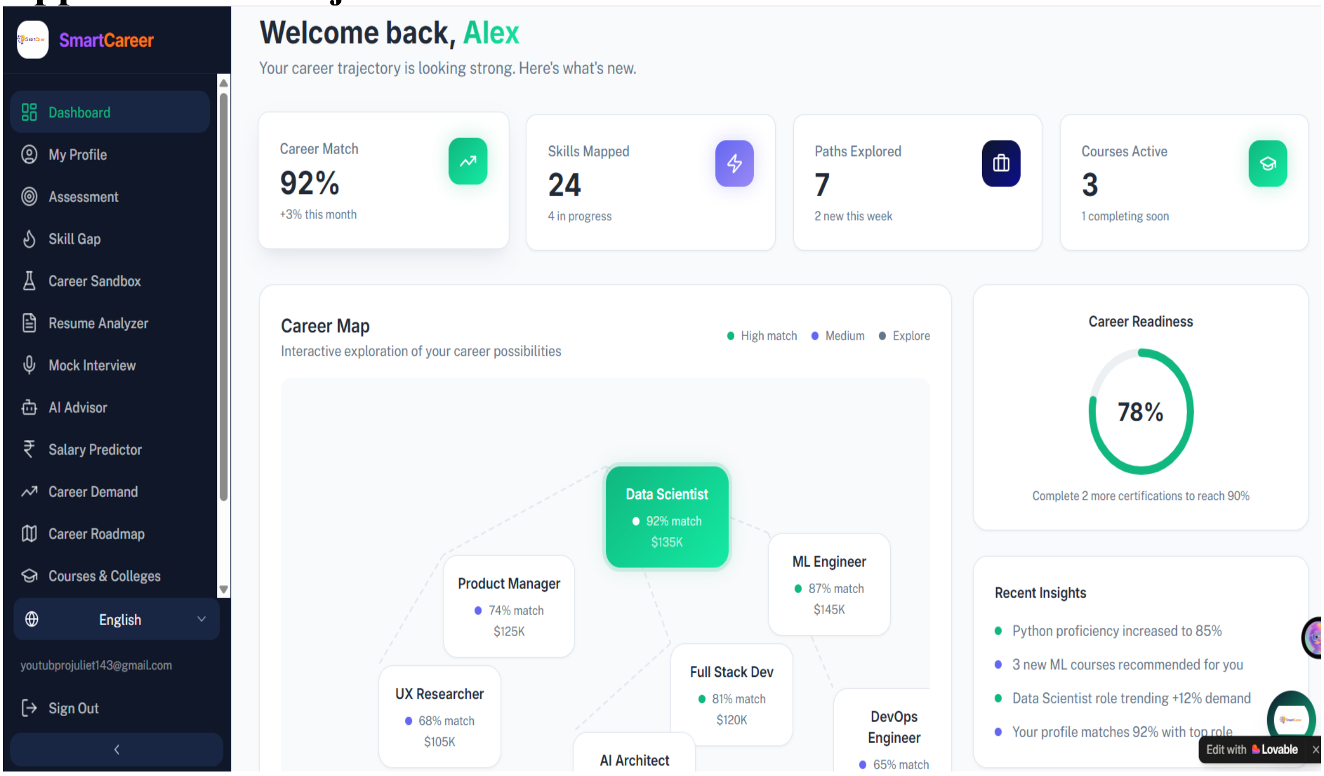
APPENDIX

Appendix A – Project Screenshots



9A.1 Project Screenshots

The following screenshots demonstrate different modules and functionalities of the proposed system **"CareerShala: One-Stop Personalized Career & Education Advisor Software"**.

Screenshots Included:

- Login and Authentication Page
- Dashboard Interface (Career Overview & Insights)
- Student Profile Management Page
- Career Assessment Module
- Skill Gap Analyzer Dashboard
- Resume Analyzer (ATS Score & Skill Extraction)
- AI Career Advisor Chatbot
- Salary Predictor Interface
- Career Sandbox Simulation Module
- Backend Server Execution
- Database Tables and Records

These screenshots provide a visual representation of the frontend interfaces, backend processing, and overall system functionality, highlighting how different modules interact to deliver personalized career guidance.

Appendix B – Source Code Snippets

The CareerShala system is developed using Python (backend), Streamlit (frontend), and machine learning libraries. The following code snippets represent important functionalities implemented in the system.

Frontend API Request (Streamlit Example)

```
import requests

response = requests.post(
    "http://localhost:8000/predict",
    json={"skills": ["Python", "ML"], "experience": 1}
)

print(response.json())
```

Backend API Route (FastAPI Example)

```
from fastapi import FastAPI

app = FastAPI()

@app.post("/predict")
def predict(data: dict):
    result = {"career": "Data Scientist", "match": "92%"}
    return result
```

Database Connection (Python – MySQL)

```
import mysql.connector

connection = mysql.connector.connect(
    host="localhost",
    user="root",
    password="password",
    database="CareerShala_db"
)
```

```
cursor = connection.cursor()
```

These code snippets illustrate API communication, backend processing, and database connectivity within the CareerShala system.

Appendix C – Database Tables

The CareerShala system uses a relational database to store user details, career data, analysis results, and system logs.

Table C.1 Database Tables

Table Name	Purpose
Users	Stores user profile information
Skills	Stores user skills and proficiency
Assessments	Stores questionnaire responses
Resume_Data	Stores parsed resume details
Career_Recommendations	Stores recommended career roles
Salary_Predictions	Stores salary estimation results
Reports	Stores generated reports
User_Activity	Tracks user interactions and history

These tables ensure structured data storage and efficient retrieval for analysis and reporting.

Appendix D – API Testing Results

API testing was performed to validate communication between frontend and backend modules of the CareerShala system. The testing ensures that data is correctly sent, processed, and returned without errors.

APIs Tested:

- User Authentication API
- Profile Management API
- Resume Analyzer API
- Skill Gap Analysis API
- Career Recommendation API
- Salary Prediction API

- AI Advisor Chat API
- Report Generation API

Testing Tools Used:

- Postman
- Browser-based API testing
- Streamlit interface testing

The testing results confirmed that all APIs function correctly, returning accurate responses and maintaining smooth communication between system components.

Appendix E – Datasets Used

The CareerShala system uses multiple datasets for training and analysis.

Datasets Included:

- Job roles and skill mapping dataset
- Salary dataset based on role, experience, and location
- Resume sample dataset for NLP processing
- Skill classification dataset

These datasets are preprocessed using data cleaning, normalization, and feature extraction techniques to improve model accuracy and performance.

Appendix F – Additional Supporting Documents

This section includes auxiliary materials that support the CareerShala project.

Documents Included:

- Project proposal and approval document
- System architecture diagrams
- Use Case Diagram
- Data Flow Diagram (DFD)
- Gantt Chart
- Test reports

These documents provide additional insights into project planning, design, and implementation.

Appendix F – Project Planning Documents

Project planning documents include the timeline, development strategy, and management activities followed during the implementation of the CareerShala system. These documents provide a structured approach to project execution and help in tracking progress, identifying milestones, and ensuring timely completion of tasks.

Included Documents:

- Gantt Chart (Project timeline and milestones)
- Agile Development Workflow
- Sprint Planning and Iterations
- Task Distribution among modules
- Development Phases (Analysis, Design, Implementation, Testing)

These documents help in understanding project scheduling, resource allocation, and systematic execution of the CareerShala system.

Appendix G – Dataset Information

The CareerShala system uses multiple datasets for training machine learning models, performing analysis, and generating recommendations. These datasets are carefully selected and preprocessed to ensure accuracy and relevance.

Dataset Usage:

- Career Recommendation and Role Matching
- Skill Gap Analysis and Mapping
- Salary Prediction based on experience, role, and location
- Resume Parsing and NLP-based Skill Extraction
- Testing and Simulation of AI Models

The datasets are used only for educational, analytical, and research purposes. Proper preprocessing techniques such as cleaning, normalization, and feature extraction are applied to improve system performance.

Appendix H – Installation and Execution Steps

The following steps describe how to install and execute the CareerShala system, including both frontend and backend components.

Frontend Execution (Streamlit Application)

pip install -r requirements.txt

streamlit run app.py

Frontend URL:
http://localhost:8501

Backend Execution (FastAPI / Python Server)

cd backend

pip install -r requirements.txt

uvicorn main:app --reload

Backend
URL: http://localhost:8000

These commands ensure proper setup and execution of the CareerShala system in a local development environment.

Appendix I – Testing Reports

Testing was conducted to validate the functionality, performance, and reliability of the CareerShala system. Multiple testing strategies were applied to ensure smooth operation across all modules.

Testing Activities:

- Functional Testing of all modules
- Integration Testing between frontend and backend
- Database Testing for storage and retrieval
- User Interface (UI) Testing for usability and responsiveness
- Performance Testing for speed and efficiency
- AI Model Testing for prediction accuracy

The test results confirmed successful execution of all major functionalities, with accurate outputs and stable system performance.

Appendix J – Additional Diagrams

The following diagrams are included as supporting materials to provide a better understanding of the system design and workflow of the CareerShala project.

Diagrams Included:

- Use Case Diagram (User interactions with system)
- Data Flow Diagram (DFD)
- System Architecture Diagram
- Functional Block Diagram
- Flowchart of System Process

These diagrams visually represent the system structure, data flow, and interaction between different modules, enhancing clarity and understanding of the project.